DAY 29 | HANDS-ON ACTIVITIES

GenAI System Evaluation: Problem Statements

Seven problems on evaluating the policy assistant. Offline metrics are simplified stand-ins; with a key, use LLM judges and real embeddings.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Write the Eval Set

Level: Warm-up | **Concepts:** dataset design

Write 12 questions with an expected key phrase, the source sentence, a topic tag and a language tag. Include 3 Japanese questions and 2 unanswerable ones.

Sample run / expected result

```
{"q": "How many sick days?", "expect": "8 days", "source": "8 days of paid sick lea
```

Hints

- Use the words real users would use, not the document's words.

Problem 2: Retrieval Metrics

Level: Easy | **Concepts:** hit rate, MRR

Compute hit@1, hit@3 and MRR for your eval set.

Sample run / expected result

```
hit@1 0.67 | hit@3 0.83 | MRR 0.74
```

Hints

- MRR = mean of 1 / rank of the first correct result.

Problem 3: Faithfulness

Level: Medium | **Concepts:** claim checking

Compute faithfulness for each answer. Then use the planted hallucinating generator and show the score drop.

Sample run / expected result

```
Faithful generator: 0.96
Hallucinating generator: 0.52
```

Hints

- Split answers into sentences; check each against the retrieved context.

Problem 4: Error Analysis

Level: Medium | **Concepts:** retrieval miss vs generation error

Classify every failure as retrieval miss or generation error and count each. Recommend the first fix.

Sample run / expected result

```
retrieval_miss: 3 (mostly Japanese)
generation_error: 1
First fix: multilingual embeddings
```

Hints

- Read the failures before choosing a fix.

Problem 5: Slice by Language and Topic

Level: Medium | **Concepts:** bias analysis

Report correctness by language and by topic. Explain any gap and how you would close it.

Sample run / expected result

```
en: 0.89 | ja: 0.00 (offline)
leave: 1.00 | travel: 0.75
```

Hints

- Offline embeddings cannot match Japanese to English text; real multilingual models can.

Problem 6: Build an LLM Judge

Level: Challenging | **Concepts:** rubric, reasons, calibration

Write a rubric-based judge (1 to 5 correctness with reasons). Label 10 answers yourself and compare: how often do you agree within 1 point?

Sample run / expected result

```
Agreement within 1 point: 8/10
Judge was lenient on partially correct answers
```

Hints

- Ask for the reason before the score.

Problem 7: Position Bias Test

Level: Stretch | **Concepts:** judge bias

Ask a judge to pick the better of two answers, then swap their order. Measure how often the choice flips.

Sample run / expected result

```
Choice flipped on swap: 3/10 -> position bias present
```

Hints

- Average both orders to reduce the bias.

Hakaru (測る): to measure